

Submission C solution to Problem 9

Abstract

Let $R_n^{(m)}$ be the diagonal coinvariant algebra of $\mathfrak{S}_n$ in m sets of n commuting variables, and write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

For hook shapes $\lambda = (a+1, 1^b)$, we give a positive ordered-set-partition interpretation of $c_{\lambda}(n)$.

1 Statement of the hook formula

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions

$$\pi = (B_1 \mid B_2 \mid \dots \mid B_k)$$

of $[r] = \{1, \dots, r\}$ into k nonempty blocks. Define

$$\text{inv}(\pi) = \#\{(i, j, x) : 1 \leq i < j \leq k, x \in B_i, x > \min(B_j)\}.$$

Also write $[k]_q = 1 + q + \dots + q^{k-1}$.

Theorem 1 (Hook coefficients). *Let $n \geq 1$. For $a, b \geq 0$, write the hook*

$$(a \mid b) := (a+1, 1^b).$$

Then

$$\sum_{a, b \geq 0} c_{(a \mid b)}(n) q^a z^b = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Equivalently,

$$c_{(a+1, 1^b)}(n) = \#\left\{(r, \pi, t) : \begin{array}{l} 1 \leq r \leq n-1, \pi \in \text{OP}_{r, r-b}, \\ 0 \leq t \leq r-b-1, \text{inv}(\pi) + t = a \end{array} \right\}.$$

In particular, $c_{(a+1, 1^b)}(n) = 0$ unless $0 \leq b \leq n-2$.

2 Ordered set partitions and q -Stirling numbers

Let

$$A_{r,k}(q) := \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Lemma 2. *The polynomials $A_{r,k}(q)$ satisfy*

$$A_{r,k}(q) = [k]_q (A_{r-1,k}(q) + A_{r-1,k-1}(q)),$$

with $A_{1,1}(q) = 1$. Hence

$$A_{r,k}(q) = [k]!_q \text{Stir}_q(r, k),$$

where $\text{Stir}_q(r, k)$ is the Carlitz q -Stirling number defined by

$$\text{Stir}_q(r, k) = [k]_q \text{Stir}_q(r-1, k) + \text{Stir}_q(r-1, k-1).$$

Proof. Remove the largest element r from an ordered set partition $\pi = (B_1 | \dots | B_k)$.

If r lies in a block B_i of size at least 2, removing r gives an ordered set partition in $\text{OP}_{r-1,k}$. Conversely, inserting r into block i of a partition in $\text{OP}_{r-1,k}$ creates exactly $k-i$ new inversions, since $r > \min(B_j)$ for every block $j > i$. Thus these insertions contribute

$$\sum_{i=1}^k q^{k-i} A_{r-1,k}(q) = [k]_q A_{r-1,k}(q).$$

If r is a singleton block, deleting that block gives a partition in $\text{OP}_{r-1,k-1}$. Conversely, inserting the singleton block $\{r\}$ in position i creates again $k-i$ new inversions. These insertions contribute $[k]_q A_{r-1,k-1}(q)$. Adding the two cases proves the recurrence. The identification with $[k]!_q \text{Stir}_q(r, k)$ follows from the same recurrence and initial condition. $\square$

3 Superspace and hook extraction

Let $\Omega_n = \mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n)$, where the x_i commute and the θ_i anticommute. Let $\mathfrak{S}_n$ act diagonally by

$$w \cdot x_i = x_{w(i)}, \quad w \cdot \theta_i = \theta_{w(i)}.$$

The superspace coinvariant ring is

$$SR_n := \Omega_n / \langle (\Omega_n)_+^{\mathfrak{S}_n} \rangle.$$

Let q track x -degree and z track θ -degree.

Lemma 3 (Hook specialization). *Let*

$$C_n(q, z) := \sum_{a,b \geq 0} c_{(a|b)}(n) q^a z^b.$$

Then

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z)C_n(q, z).$$

Proof. The construction of diagonal coinvariants is functorial in the row-space. For a purely even row-space of dimension m , its character is

$$\sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

Evaluating the same polynomial functor on a $1|1$ -dimensional superspace gives the superspace coinvariant ring SR_n . By the Berele–Regev hook Schur character theory, the ordinary Hilbert

character of the Schur functor $\mathbb{S}_\lambda$ on one even variable q and one odd variable z is zero unless λ is a hook. Moreover,

$$hs_{(a|b)}(q; z) = q^{a+1}z^b + q^a z^{b+1} = (q+z)q^a z^b,$$

and $hs_\emptyset(q; z) = 1$. Therefore

$$\text{Hilb}(SR_n; q, z) = 1 + \sum_{a,b \geq 0} c_{(a|b)}(n)(q+z)q^a z^b = 1 + (q+z)C_n(q, z).$$

This is the standard hook Schur specialization; see Berele–Regev [1] or Macdonald [3, Ch. I, §3, Ex. 23]. $\square$

4 Hilbert series of SR_n

Rhoades and Wilson proved the following Hilbert series formula for the superspace coinvariant ring [4, Theorem 1.1 and Eq. (1.10)]:

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} [k]_q! \text{Stir}_q(n, k).$$

Using the preceding lemma, this can be written as

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Set

$$H_n(q, z) := \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Then $H_1(q, z) = 1$.

Lemma 4. *For $n \geq 2$,*

$$H_n(q, z) - H_{n-1}(q, z) = (q+z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}(q).$$

Consequently,

$$\frac{H_n(q, z) - 1}{q+z} = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q).$$

Proof. Using the recurrence from the ordered-set-partition lemma,

$$A_{n,k} = [k]_q (A_{n-1,k} + A_{n-1,k-1}).$$

Therefore

$$\begin{aligned} H_n &= \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k-1} \\ &= \sum_{k=1}^{n-1} z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^{n-1} z^{n-1-k} [k+1]_q A_{n-1,k} \\ &= \sum_{k=1}^{n-1} z^{n-1-k} (z[k]_q + [k+1]_q) A_{n-1,k}. \end{aligned}$$

Since

$$z[k]_q + [k+1]_q = z[k]_q + 1 + q[k]_q = 1 + (q+z)[k]_q,$$

we obtain

$$H_n = H_{n-1} + (q+z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}.$$

Telescoping from $H_1 = 1$ proves the second formula. $\square$

5 Proof of the theorem

By the hook-specialization lemma and the Rhoades–Wilson Hilbert series formula,

$$1 + (q+z)C_n(q, z) = H_n(q, z).$$

Thus

$$C_n(q, z) = \frac{H_n(q, z) - 1}{q+z}.$$

The preceding lemma gives

$$C_n(q, z) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Since

$$[k]_q = \sum_{t=0}^{k-1} q^t,$$

the coefficient of $q^a z^b$ counts triples

$$(r, \pi, t)$$

such that $1 \leq r \leq n-1$, $\pi \in \text{OP}_{r,k}$, $k = r-b$, $0 \leq t \leq k-1$, and

$$\text{inv}(\pi) + t = a.$$

This is precisely the stated interpretation of $c_{(a+1, 1^b)}(n)$.

References

- [1] A. Berele and A. Regev, Hook Young diagrams with applications to combinatorics and to representations of Lie superalgebras, *Adv. Math.* **64** (1987), 118–175.
- [2] F. Bergeron, Multivariate diagonal coinvariant spaces for complex reflection groups, *Adv. Math.* **239** (2013), 97–108.
- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.
- [4] B. Rhoades and A. T. Wilson, The Hilbert series of the superspace coinvariant ring, *Forum Math. Pi* **12** (2024), Paper No. e16.